Cyprien Dcunha

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Experience

Amazon Dec 2021 – Nov 2025
Software Development Engineer 2 (Finance Automation) *Hyderabad*

- Drove VendorHoldbacks re-architecture from monolith to 11 microservices, leading a team of 3 developers, enabling 6M transactions (worth $2.1B) at 2000 TPS. Transformed batch processing to event-driven architecture, slashing processing time from 24h to near real-time and reducing vendor inquiries by 23.5% across 14+ countries.
- Engineered an LLM-driven Financial Reconciliation Engine processing up to 10M records per day, accelerating anomaly detection and reducing manual work by 65%. Utilized RAG against historical customer notes and LLM-based semantic mapping to achieve a recon accuracy of 67%.
- Architected a high-throughput Rule Engine processing 5,000+ concurrent business rules (50ms latency, 2,000 TPS). Introduced self-service portal cutting rule creation time from 4 weeks to minutes, incorporating versatile business logic via Drools framework.
- Participated extensively in technical design reviews, providing feedback focused on improving system architecture, scalability, and identifying potential anti-patterns.
- Successfully delivered multiple projects that enhanced vendor experience by owning high and low level design and implementation, also leading to a decrease in Amazon's debit balance by up to $68MM.
- Conducted over 1,100 code reviews, focusing on enhancing code quality, maintainability, performance, and adherence to best practices.
- Spearheaded roadmap planning by advising senior leadership on technical feasibility and project complexity, enabling strategic prioritization and goal setting.

Amazon May 2019 – Dec 2021
Software Development Engineer 1 (Finance Automation) *Hyderabad*

- Developed ML-driven automated holdback system preventing $14.7MM in debit balance exposure. Built scalable pipeline processing 90,000+ vendors' historical data with configurable risk engine using linear regression for seasonal patterns, eliminating manual assessment for 3,600+ high-risk vendors during peak events (Prime Day, Black Friday).
- Implemented a data segregation system for special merchants, restricting data access across over 20 input sources and ensuring merchant-specific information security.
- Slashed support tickets from 1200+ (2020) to under 20 (2022) through systematic issue resolution. Established robust on-call processes for high-severity incidents, significantly reducing operational load.

IRageCapital Advisory Pvt. Ltd. Jun 2017 – Aug 2017
C++ Development Intern (High frequency Trading Platform) *Mumbai*

- Developed a high-performance market replay simulator in C++, achieving less than 50 microseconds latency per market execution, enabling backtesting of trading strategies against historical market data.

Education

Dwarkadas J. Sanghvi College of Engineering Jun 2015 – May 2019
Bachelor of Engineering, Computer Science (CGPA: 9.08 / 10) *Mumbai*

Awards and Recognitions

- Earned Amazon SDE Recognition Awards (2021, 2022) for consistently raising the engineering bar.
- Finalist in Amazon FinAuto Hackathon 2023 for developing IDE tooling for AWS Step Functions development.
- Winner of IBM's Call for Code Challenge at AngelHack, securing $1,000 prize.
- Secured 3rd place and $1,100 prize in TopCoder Development Challenge.

Technical Skills

Languages & Core: Java, Python, C++, JavaScript, TypeScript, MySQL, Data Structures, Algorithms, CI/CD

AWS & Cloud: Lambda, ECS, Step Functions, EventBridge, SQS/SNS, DynamoDB, RDS, S3, Redshift

Gen-AI: LLMs, Prompt Engineering, AWS Bedrock, AWS AgentCore, RAG, Agent Memory, Laminar, LangChain

Architecture: Distributed Systems, Agentic Systems, Microservices, Event-Driven, Domain-Driven, REST APIs

Frameworks & Tools: Apache Commons, Guice, Dagger, Jackson, JUnit, Mockito, Maven, Gradle, Drools, NumPy, Pandas